signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

signal.sci

```
clc;
clear;
// Time vector (0 to 5 sec, step 0.001 s)
t = 0:0.001:5;
dt = t(2) - t(1); // time step
// Define continuous-time signal x(t) = e^(-2t) * sin(3t)
x = exp(-2*t) .* sin(3*t);
// s-domain values
sigma = 0; // Damping factor
omega = -20:0.5:20; // Frequency range
// Initialize Laplace magnitude array
X_mag = zeros(1, length(omega));
// Numerical Laplace calculation (manual trapezoidal rule)
for k = 1:length(omega)
s = sigma + %i*omega(k);
Xs = x .* exp(-s*t);
// trapezoidal integration
X_mag(k) = abs( sum(Xs(1:$-1) + Xs(2:$)) * dt / 2 );
end
// Plot magnitude spectrum
clf;
plot(omega, X_mag, 'b');
xlabel("Frequency (rad/s)");
ylabel("|X(s)|");
title("Numerical Laplace Transform Magnitude");
```

Graphic window number 0

File Tools Edit ?

Graphic window number 0 ?

